

Project Proposal NLP

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1 Brief Introduction

Recent work [1] shows that with enough demonstrations, LLMs can reorganize concept representations to reflect *context-specified structure* (e.g., grid/ring graphs), measurable via decreasing Dirichlet energy and increasing rule-following accuracy, *i.e.*, sum of probability mass assigned to valid continuations of sequences. Our work will have two goals: (i) extend their work by experimenting on overlooked questions, (ii) analyze the representations produced by hierarchical graphical structures and structures inducted by a grammar.

Specifically, we want to answer the following questions: What is the relationship between phrase structures and distances in the representation space? Are we able to detect *hierarchical* geometries induced in-context by simple trees or grammars? Lastly, how much the learned geometries are *memory* dependent?

2 Planned experiments

2.1 Extensions of the original setup

Token design (reduce semantic confounds). Use semantically uncorrelated terminals (synthetic or selected to have low mutual predictability). This focuses the analysis on structure rather than lexical semantics.

Frequency control (torus grid). Repeat the grid-walk experiment with periodic boundary conditions to remove boundary-driven visitation-frequency artifacts.

Transition probabilities / geometry relationship. Vary edge sampling frequencies (e.g., make a subset of edges 10× more likely while keeping the graph connected). Test whether higher transition probability implies shorter representation distance and lower weighted Dirichlet energy.

Generalization beyond memorization (held-out node). Hold out a node k from context. Provide contexts ending at neighbors of k and test: (i) whether

the model predicts k , and (ii) whether k 's representation (obtained via probes) lies near the location implied by the learned geometry.

Days of the week (grid structure). Repeat the same experiment with “days of the week” using a different parsing structure, in order to avoid potential bias induced by the circle symmetry in the author’s work.

2.2 ICL from hierarchical structure

Binary tree (one token per node). Generate sequences by traversals over a directed binary tree. Test whether representations reflect tree distance / depth (e.g., depth decodability using linear probes, Dirichlet energy on tree edges vs. context length).

Binary tree (multiple tokens per node). Assign 3–5 representative tokens per node (same structural role, not semantic similarity). Compare within-node variance vs. between-node distances. If the model learns node-level categories, tokens from the same node should cluster and node centroids should recover the tree geometry.

PCFGs (constituents vs. paths). PCFGs introduce constituent structure (e.g., NP vs. VP) and allow ambiguity/recursion, which cannot be reduced to a single path (as in trees). Starting for example from:

$$\begin{aligned} S &\rightarrow NP VP \text{ (1.0)} \\ NP &\rightarrow Det N \text{ (0.7)} \mid N \text{ (0.3)} \\ VP &\rightarrow V NP \text{ (1.0)} \end{aligned}$$

with semantically neutral terminals. Test: (i) rule learning and rule probability learning (e.g., after “Det N V” predict *Det* vs. *N* with or without probabilities); (ii) category induction via novel terminals inserted in N positions (i.e. can the model assign the right category to novel terminals by moving them towards the category cluster?).

2.3 Representation analysis

Across settings, measure alignment between representation geometry and latent structure using: (i) hierarchical clustering / dendrograms; (ii) Dirichlet energy (and weighted variants); (iii) linear probes for structural labels (tree depth; PCFG non-terminal class). Tracking these quantities vs. context size N may allow to identify possible phase-transition-like changes in ICL dynamics.

3 Estimate of computational resources

They experiment on the internal representations of the following base models: Llama3.1-8B; Gemma-2-2B / Gemma-2-9B.

References

- [1] Core Francisco Park, Andrew Lee, Ekdeep Singh Lubana, Yongyi Yang, Maya Okawa, Kento Nishi, Martin Wattenberg, and Hidenori Tanaka. Iclr: In-context learning of representations. *arXiv preprint arXiv:2501.00070*, 2024.